

Cory Morgan

AWS Certified Solutions Architect – Associate

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GitHub | LinkedIn | Portfolio

Summary

AWS Certified Solutions Architect with hands-on experience building cloud-native and AI/ML solutions using AWS, Terraform, Python, and JavaScript. Strong at translating business needs into scalable architectures, explaining technical tradeoffs to technical and non-technical stakeholders, and building production-minded projects across serverless, DevOps, security, storage, and machine learning domains.

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL, Node.js

Cloud: AWS Lambda, S3, DynamoDB, API Gateway, Route 53, CloudFront, EC2, IAM, VPC, CloudWatch, Systems Manager

Architecture/DevOps: Terraform, Docker, GitHub Actions, CI/CD, Infrastructure as Code, Monitoring, Cost Optimization

Architecture Domains: Serverless Architecture, Cloud Security, Storage, DevOps, Application Development, Machine Learning

AI/ML: SageMaker, TensorFlow, PyTorch, Hugging Face, Gemma LLM, Prompt Engineering, Model Evaluation, OpenCV, YOLO

Web/Tools: React, Express, PostgreSQL, Git, GitHub

Certifications

- AWS Certified Solutions Architect – Associate, AWS, May 2025
- AI Engineering Fellowship, KaggleX, Dec 2024
- Semantic Segmentation with Amazon SageMaker, Coursera, Feb 2025
- Computer Vision & Deep Learning with PyTorch, Udemy, Jan 2025

Projects

AWS Cloud Resume Challenge

- Architected and deployed a serverless AWS application using S3, CloudFront, Route 53, Lambda, API Gateway, and DynamoDB for low-cost hosting and minimal infrastructure management.
- Provisioned infrastructure with Terraform and automated deployments through GitHub Actions CI/CD to enable repeatable releases and reduce manual configuration drift.
- Implemented monitoring with CloudWatch and designed for reliability, operational simplicity, and a monthly cost target under \$10.

AI-Powered Semantic FAQ Search Platform

- Designed and deployed a secure AWS architecture using VPC, EC2, S3, IAM, Docker, and Terraform to host a containerized search platform.
- Built hybrid BM25 plus vector search workflows with embeddings to improve relevance and retrieval quality.
- Balanced usability, scalability, and cost while documenting technical tradeoffs and deployment decisions.

Cloud VPN with Ad-Blocking

- Deployed a WireGuard VPN on AWS EC2 with Terraform managing 10 infrastructure resources.
- Implemented Systems Manager Session Manager for secure administration without SSH key exposure.
- Applied security and cost-conscious design choices to keep the environment lightweight and manageable within AWS free-tier limits.

Video Instance Segmentation Model

- Engineered a YOLOv8-based computer vision pipeline for real-time vehicle detection and tracking.
- Improved model quality with custom preprocessing, augmentation, annotation, and fine-tuning workflows.
- Exported to ONNX for efficient inference and production-minded deployment flexibility.

Monarch: AI-Powered Personal Finance App

- Built backend services with Node.js, Express, and PostgreSQL for an AI-enabled finance application.
- Integrated chatbot capabilities and tested outputs for clarity, relevance, and safe user guidance.
- Delivered features through iterative Agile sprints with version control and CI/CD practices.

Experience

AI Engineering Fellow – KaggleX

Aug 2024 – Dec 2024 | Remote

- Developed and refined an AI medical chatbot using Gemma LLM and TensorFlow.
- Evaluated outputs for factual accuracy, safety, and usability while iterating on prompts and model behavior.
- Documented issues, tested edge cases, and communicated findings clearly in a collaborative project environment.

Lead Care Coordinator – Spear Physical Therapy

Jun 2023 – Present | Brooklyn, NY

- Lead cross-functional coordination across patients, providers, and internal teams in a high-volume clinic environment.
- Translate complex insurance, billing, and scheduling issues into clear next steps and workflows for varied stakeholders.
- Support workflow improvements, issue resolution, and day-to-day operations requiring strong judgment, prioritization, and communication.

Software Engineering Student – Thinkful, Chegg Skills Inc.

Apr 2023 – Oct 2023 | Remote

- Completed an intensive full-stack software engineering program focused on practical application development.
- Built responsive web applications with JavaScript and modern development workflows.

Education

Rochester Institute of Technology – B.S. in Business Management & Marketing